

For Packet Size = 2320:

```
student@student-26:~/Downloads/ns-allinone-3.40/ns-3.40$ ./ns3 run scratch/first.cc
[0/2] Re-checking globbed directories...
ninja: no work to do.
At time +2s client sent 2320 bytes to 10.1.1.2 port 9
At time +2.0058s server received 2320 bytes from 10.1.1.1 port 49153
At time +2.0058s server sent 2320 bytes to 10.1.1.1 port 49153
At time +2.01159s client received 2320 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
  Tx Bytes: 2348
  Rx Bytes: 2348
  Mean Delay: 0.0057952 s
  Throughput: 1043.56 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
  Tx Bytes: 2348
  Rx Bytes: 2348
  Mean Delay: 0.0057952 s
  Throughput: 1043.56 bps
student@student-26:~/Downloads/ns-allinone-3.40/ns-3.40$ |
```

For Packet Size = 1644:

```
student@student-26:~/Downloads/ns-allinone-3.40/ns-3.40$ ./ns3 run scratch/first.cc
[0/2] Re-checking globbed directories...
ninja: no work to do.
At time +2s client sent 1644 bytes to 10.1.1.2 port 9
At time +2.00471s server received 1644 bytes from 10.1.1.1 port 49153
At time +2.00471s server sent 1644 bytes to 10.1.1.1 port 49153
At time +2.00943s client received 1644 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
  Tx Bytes: 1672
  Rx Bytes: 1672
  Mean Delay: 0.0047136 s
  Throughput: 743.111 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
  Tx Bytes: 1672
  Rx Bytes: 1672
  Mean Delay: 0.0047136 s
  Throughput: 743.111 bps
student@student-26:~/Downloads/ns-allinone-3.40/ns-3.40$ |
```

For Packet Size = 232:

```

student@student-26:~/Downloads/ns-allinone-3.40/ns-3.40$ ./ns3 run scratch/first.cc
[0/2] Re-checking globbed directories...
ninja: no work to do.
At time +2s client sent 232 bytes to 10.1.1.2 port 9
At time +2.00242s server received 232 bytes from 10.1.1.1 port 49153
At time +2.00242s server sent 232 bytes to 10.1.1.1 port 49153
At time +2.00484s client received 232 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
  Tx Bytes: 260
  Rx Bytes: 260
  Mean Delay: 0.0024192 s
  Throughput: 115.556 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
  Tx Bytes: 260
  Rx Bytes: 260
  Mean Delay: 0.0024192 s
  Throughput: 115.556 bps
student@student-26:~/Downloads/ns-allinone-3.40/ns-3.40$ |

```

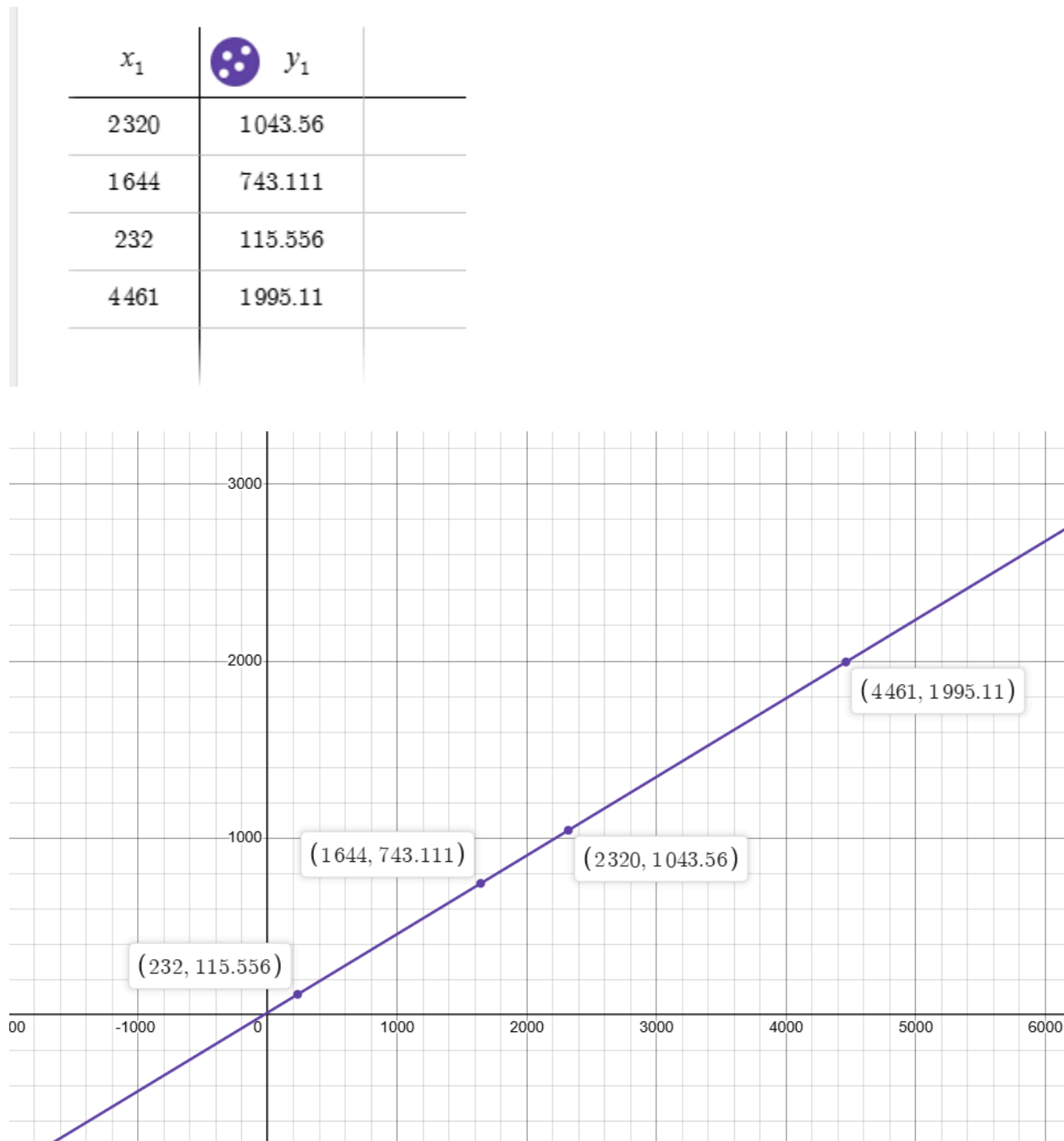
For Packet Size = 4461:

```

student@student-26:~/Downloads/ns-allinone-3.40/ns-3.40$ ./ns3 run scratch/first.cc
[0/2] Re-checking globbed directories...
ninja: no work to do.
At time +2s client sent 4461 bytes to 10.1.1.2 port 9
At time +2.00929s server received 4461 bytes from 10.1.1.1 port 49153
At time +2.00929s server sent 4461 bytes to 10.1.1.1 port 49153
At time +2.01858s client received 4461 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
  Tx Bytes: 4489
  Rx Bytes: 4489
  Mean Delay: 0.0092912 s
  Throughput: 1995.11 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
  Tx Bytes: 4489
  Rx Bytes: 4489
  Mean Delay: 0.0092912 s
  Throughput: 1995.11 bps
student@student-26:~/Downloads/ns-allinone-3.40/ns-3.40$ |

```

Graph:



The graph shows a clear pattern: as the packet size increases, the throughput increases as well. The four points line up almost perfectly straight, which tells us the two grow together in a steady, proportional way. In other words, when the packet size roughly doubles, the throughput roughly doubles too. The reason for this is simple. Throughput here is just the amount of data (in bits) sent over a fixed period of time. Since the simulation sends only one packet within the same time window each time, a bigger packet means more bits are pushed through in that window, and so

the throughput goes up. And it keeps rising smoothly with the packet size. That's why we get this nice, nearly straight upward line.